

1. **List 5 common BIOS or POST error messages you might see when starting a PC and write what hardware or configuration issue each one usually points to.**

1. "CMOS Checksum Error – Defaults Loaded"

Usually points to:

- Weak or dead CMOS battery (motherboard battery).
- BIOS settings have been lost or corrupted.
- Incorrect BIOS configuration.

2. "Keyboard Error or No Keyboard Present"

Usually points to:

- Keyboard not connected properly.
- Faulty keyboard cable or USB/PS2 port.
- Defective keyboard.

3. "No Boot Device Found" / "Disk Boot Failure"

Usually points to:

- Hard drive or SSD not detected.
- Loose SATA or power cable.
- Incorrect boot order in BIOS.
- Failed storage device.

4. "Memory Test Failed" / "Memory Error"

Usually points to:

- Faulty RAM module.
- RAM not seated correctly.
- Incompatible memory settings.

5. "CPU Fan Error"

Usually points to:

- CPU cooling fan not connected to the correct motherboard header.
- Fan has stopped working.
- Fan speed is below the BIOS warning threshold.

2. **Given this scenario: On boot, a PC beeps 3 times in a repeating pattern and does not display anything on the screen. Research and write what this POST beep code typically means for an AMI BIOS, and suggest two troubleshooting steps.**

Scenario

- **Problem:** When the PC is turned on, it beeps **3 times in a repeating pattern** and nothing appears on the monitor.

AMI BIOS POST Beep Code

- For a typical **AMI BIOS**, **3 short beeps** usually indicate a **memory (RAM) error**. This means the BIOS cannot detect or properly use the system memory during the Power-On Self-Test (POST).

Two Troubleshooting Steps

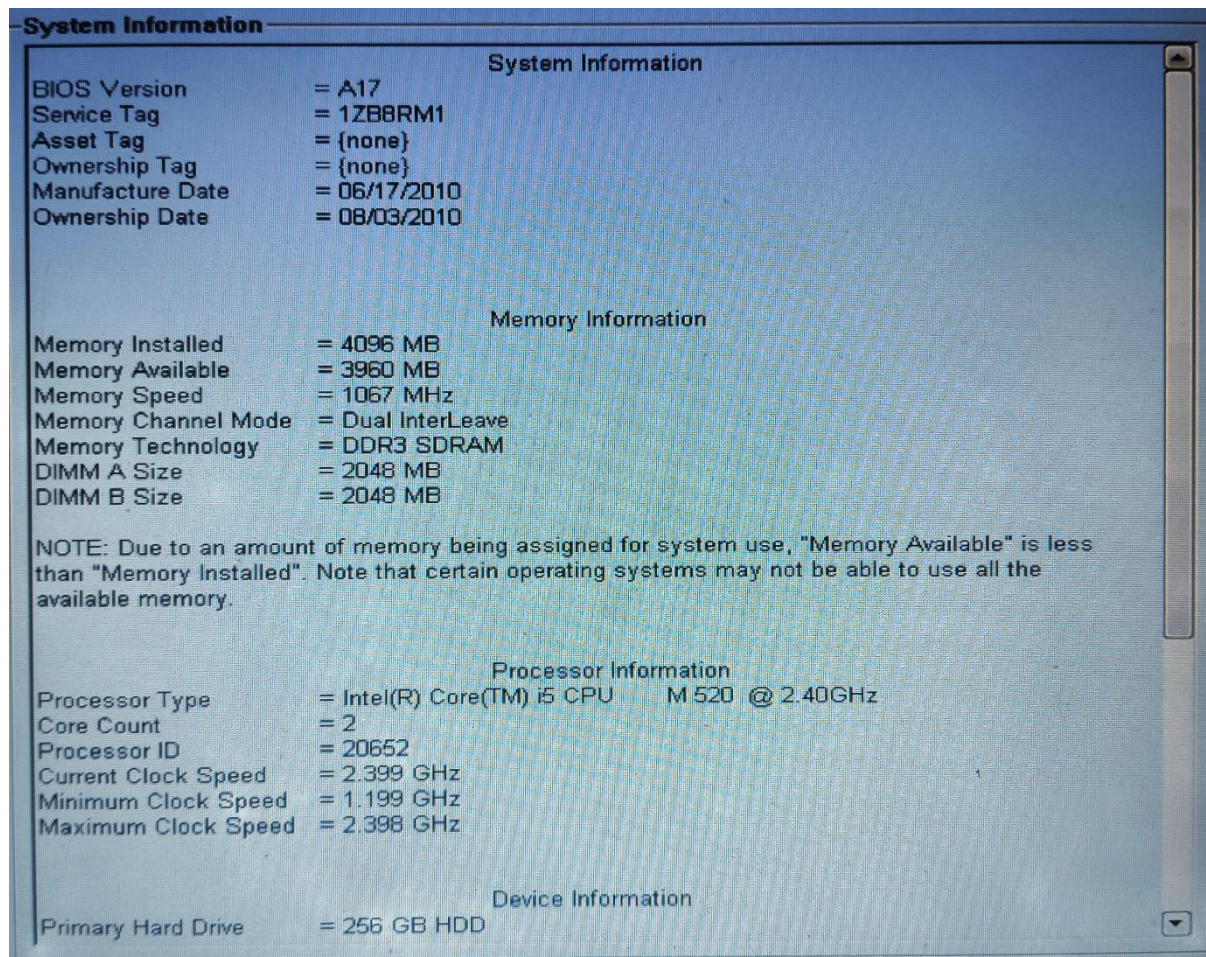
1. Reseat the RAM

- Turn off the PC and unplug the power.
- Remove the RAM module(s) and reinstall them firmly into the memory slots.
- Make sure they click securely into place.

2. Test the RAM

- Try booting the PC with **one RAM stick at a time** if multiple modules are installed.
- If available, test with a known working RAM module or try a different memory slot to determine whether the RAM or motherboard slot is faulty.

3. **Access the BIOS setup utility on any available PC or virtual machine, take a screenshot of the main BIOS screen, and identify the motherboard manufacturer and BIOS version.**



4. You are given a screenshot of a POST error message that says 'CMOS checksum error – Defaults loaded.' Explain what this means and describe how you would resolve this issue in a real system.

How to Resolve This Issue

Fixing this issue typically requires a straightforward hardware check and system reconfiguration

1. **Replace the CMOS Battery:** This is the most common fix. Unplug the computer, open the case, and locate the silver, coin-sized **CR2032 battery** on your motherboard. Pop it out and replace it with a fresh one to restore power to the CMOS chip.
2. **Access the BIOS/UEFI:** Restart your computer and press the required key (usually **Delete, F2, F10, or Esc**) to enter the BIOS menu.
3. **Reset the Time & Settings:** Because the defaults were loaded, your system date and time are likely incorrect. Set the current time/date and configure any specific hardware settings (such as enabling XMP for RAM or adjusting your boot order).
4. **Save and Exit:** Select **Save Changes and Exit** (often mapped to the **F10** key).